Verification Status Note

This verification has been formally marked as Unsuccessful due to outstanding evidence that remains under external review. As of the date of this note, the issue is still under review at KDHE and therefore cannot be resolved within the current verification cycle.

Should the required evidence become available and satisfactorily address the outstanding items, the project may resubmit this verification for reassessment in accordance with Isometric's standard procedures.

This note is provided for clarity and should be read in conjunction with the full Unsuccessful Verification report that follows.

Charm Industrial Isometric Standard Project Verification Report

Client	Isometric
Company / GHG Removal Supplier	Charm Industrial
Facility or Project Name	Charm Kansas Bio-Oil Geologic Storage
Facility Address	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses, Port-Cartier, QC G0W 1X0, Canada Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 Bio-oil Injection: Vaulted Deep 7513 KS-14, Hutchinson, KS 67501
Greenhouse Gas Statement ID	Isometric ggs_1JAB40DHS1S09RW6
Net Greenhouse Gas Removals	236.23 tonnes
Removal Method	Bio-Oil Geological Storage
Removal Period	May 1 – May 31, 2024
Verification Opinion	Does NOT fully meet criteria
Verifier/Validation and Verification Body	Tim Hansen 350Solutions, Inc.

ISSUED: JAN 21, 2025

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ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Lily Schacht Kelly Inder-Nesbitt
Removal Method: Bio-Oil Geological Storage		
Report Date: January 21, 2025		
Document No: 350VR-CHARM-ISO2405-052024		
Rev: Final v 1.2		

** primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. CHARM TECHNOLOGY VERIFICATION SUMMARY

Title	Charm Kansas Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses, Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Vaulted Deep 7513 KS-14, Hutchinson, KS 67501 37.966035, -97.947845
Reporting Period	May 1- May 31, 2024
Greenhouse Gas Statement ID	Isometric ggs_1JAB40DHS1S09RW6
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (mill waste supplier) Geologic Storage Management: Vaulted Deep Regulatory Agency (Injections / Storage): Kansas Department of Health and Environment (KDHE)
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	236.23 tonnes
Removal Method	Bio-Oil Geological Storage in Salt Caverns
Verification Report Date	January 21, 2025
Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of CDR by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.2 • Bio-Oil Geological Storage Protocol v1.0.1 • Biomass Feedstock Accounting Module v1.1 • Biomass or Bio-oil Storage in Salt Caverns v1.0.2 • Embodied Emissions Accounting v 1.0.2 • Energy Use Accounting v 1.1 • Transportation Emissions Accounting v 1.0

1.1. VERIFICATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for reporting period May 1- May 31, 2024. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast and issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. The validation remains valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation).

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.4.0, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2.1*, and *Bio-oil Storage in Salt Caverns Module v1.1*.

350Solutions affirms that Charm has the majority of appropriate equipment, procedures, and protocols in place and documented to quantify GHG removal through the injection of bio-oil into salt caverns via permitted Class V wells in accordance with the requirements of the applicable Isometric standards, protocols and modules. ***However, for the reporting period of interest, the Charm injections were not compliant with a Class V well permit requirement for total chloride concentration of injectate to be at or above 190,000 mg/l. Therefore, although the values of injected carbon were verified, the project was not in compliance during the reporting period and is therefore not eligible for crediting in accordance with the applicable Isometric standards and protocols***.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD MAY, 2024

Removal ID	Supplier reference ID	Date removal started	Date removal completed	Discounted CO_2e net removed, rounded (kg)
rmv_1J9M2Z5QQ1S06YDZ	732	2024-05-10	2024-05-10	28,120
rmv_1J9SF796D1S00FB4	734	2024-05-21	2024-05-21	470*
rmv_1J9SN7T5R1S0D4D6	741	2024-05-11	2024-05-11	33,700
rmv_1J9SNCD3V1S08AK7	733	2024-05-16	2024-05-16	23,090
rmv_1J9SNJ00M1S04WV0	735	2024-05-23	2024-05-23	30,790
rmv_1J9SPS33N1S0AWM9	740	2024-05-31	2024-05-31	19,090
rmv_1J9W691T41S07MGM	736	2024-05-24	2024-05-24	18,280
rmv_1J9W7221V1S0S0EX	738	2024-05-28	2024-05-28	17,190
rmv_1J9XM7NHH1S0P8QJ	737	2024-05-25	2024-05-25	36,660
rmv_1J9XMAJFX1S0APY0	739	2024-05-30	2024-05-30	28,840
TOTAL REMOVALS				**236,230 kg (236.23 t)**

**Note that removal 734 has significantly lower net removals because Charm allocates all site and amortized (embodied) emissions to a single removal batch. This does not impact the net total removals for the reporting period.*

The quantity of 236.23 tonnes of net carbon removals has been verified with a reasonable level of assurance. However, as a result of the non-compliant chloride concentrations of injectate during the

reporting period, the Charm removals for this reporting period do not qualify for crediting under the referenced Isometric Standard and Protocol.

Verifier Information		
Verification Body	**Lead Verifier**	**Verification ID No.**
350Solutions, Inc.	Tim Hansen	VS-Charm-ISO2405

Signed: Tim Hansen (Lead Verifier) Lily Schacht (Peer Reviewer)

2. VERIFICATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Kansas Bio-oil Geologic Storage facility for the time period May 1 – May 31, 2024. The verification was conducted following the specifications of the Isometric Standard v1.4.0 (August 12, 2024), Bio-Oil Geological Storage Protocol v1.1 (September 13, 2024), Biomass Feedstock Accounting Module v1.2.1 (May 20, 2024), Biomass or Bio-oil Storage in Salt Caverns v1.1 (September 13, 2024). The verification is based on the provision and fair presentation by Charm Industrial and Isometric of a complete and accurate GHG Statement in accordance with the criteria (Table 1, Section 3.2) which is confirmed and verified by 350Solutions based on supporting evidence. The verification was conducted with a reasonable level of assurance and discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.2. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020 for completion of ISO 14034 Technology Verifications. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 20 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. Complete verifier qualifications are attached as Appendix 4.

2.3. AUDIT PROCESS FOLLOWED

The specific verification activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. A summary of the detailed results of the audit based on review of files, recalculation of removals data from raw data, confirmation of supporting evidence, and conversations with Charm staff, are provided in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes requirements from the Isometric standard, Bio-oil geologic Storage protocol, and referenced modules.

TABLE 3. VERIFICATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
November 1, 2024	Review of reporting period modifications to PDD	- Review of updated PDD and GHG Statement	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx
November 4 – 8, 2024	Review of current Validation and GHG Statement Applicability	- Review of Validation Report - Review of prior Verification Report - Identification of major and minor process steps, measurements, and calculations - Review of Standards/Protocols/Modules required for Verification - Generation of a checklist of verification issues, recommendations, and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 Charm Verification Report 2024-07-26 V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx [External] May GHG Statement Report v1.1.docx Charm May 24 GHG Notes from Charm.pdf V1.4.0, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1
November 4 – 8, 2024	Verification of Removals	- Verify changes to the system - Review of measurements and calculations - Review of durability/permit requirement monitoring data - Review of GHG statement - Identification of initial findings and follow-up request for information	File List provided in Appendix 1. Initial Findings in Appendix 2.
November 11, 2024	Verification of Removals	- Review of response to Request for Information and Initial Findings - Revision of report based on response to findings	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/ Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. Bio-oil is shipped by tanker truck or rail car to a Charm facility in El Dorado Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and liquid caustic soda (LCS) to adjust the pH of the bio-oil and the salt content or conductivity to meet the requirements of the injection facility permit for storage in salt caverns as specified by the Kansas Department of Health and Environment (KDHE) and injection partner Vaulted Deep's Class V injection permit. Following treatment, samples are taken for CHN analysis (carbon content), total acid number (TAN), and density testing and the oil is transported to the Vaulted Deep injection facility.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, including measurement of pH and conductivity both within allowable permitted ranges, the oil is trucked to Vaulted Deep's Great Plains Facility in Hutchinson, KS for injection into an underground salt

cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored on site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are detailed in their own PDD, which has been previously validated [1].

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.2
- Bio-Oil Geological Storage Protocol v1.0.1
- Biomass Feedstock Accounting Module v1.1
- Biomass or Bio-oil Storage in Salt Caverns v1.0.2
- Embodied Emissions Accounting v 1.0.2
- Energy Use Accounting v 1.1
- Transportation Emissions Accounting v 1.0

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste, converts it to bio-oil, which they pretreat and inject into salt caverns. The mixed woody waste was assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and was found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in underground salt caverns makes both the Bio-oil Geological Storage Protocol and Biomass/Bio-oil Storage in Salt Caverns Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply wood waste to the biomass processing site (AECN), the conversion of the biomass to bio-oil via AECN's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado Kansas facility, completion of pretreatment, shipment to the Vaulted Deep Great Plains facility, and injection of the bio-oil in the salt caverns. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made minor modifications to the process which have been identified in the GHGH statement and supporting documentation. The following are the only significant modifications in project activities aside from updates to delivery data and monitoring data for the reporting period:

- Charm utilized rail transport for several of the removal batches. Bio-oil was transported from AECN to rail facilities in Kansas, then offloaded and delivered by truck to Charm El Dorado, KS facility. All emissions factors, rail car fill weights (mass of bio-oil), and transportation distances for transportation by rail have been updated. For batches where tanker truck transport was used, no changes occurred. Transportation mode is identified in the GHG statement data on the Isometric dMRV platform and in supporting evidence.

- Charm purchased 275-gallon tote tanks and made minor upgrades to piping and manifold systems at its pretreatment facility in El Dorado, KS. All materials used and all tote tank materials have been accounted for in the embodied emissions for the equipment, which are allocated to removals appropriately and in accordance with the Isometric protocol requirements.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil by Charm during the period May 1 – May 31, 2024. The reporting period is based on the actual injection dates, with bio-oil production, transportation, monitoring, and related activities potentially occurring outside of this reporting period.

As stated by Charm in an addendum to its GHG Statement, there are two exceptions in the reporting period, where lots were partially consumed during this reporting period and partially consumed during previous and subsequent periods

- 041T1R0 was partially consumed in an April removal. The remaining mass was consumed in May
- 070T1R0 was partially consumed in May, the remaining mass will be consumed in June

In each case, the total mass of each origin is consumed, and all emissions are propagated downstream as confirmed in this verification.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024 and continues for five years, ending February 28, 2029. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed. Confirmation of the GHG statement has been based on the following efforts:

- Duplication of calculations of total bio-oil and CO2 equivalent injected for Removals 732 and 739 (supplier reference IDs), including:
    - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables (salt and LCS)
    - Review and calculation of average carbon content from raw data (lab reports)
    - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado KS
    - Confirmation of weigh scale raw data for deliveries
    - Re-calculation of transportation associated emissions for such transport.
    - Confirmation of raw data for fuel usage at Charm pre-treatment facility

    - Confirmation of electricity usage at Charm pretreatment facility and Vaulted Deep injection facility from raw data
    - Confirmation of electricity emissions factors for electricity for the Kansas region and recalculation of the emissions and confirmation
- Review of monitoring data as required for salt cavern injection / emplacement and post injection monitoring to ensure ongoing compliance with Class V permit

Some small discrepancies were identified based on the recalculation of net removals for the specific IDs 732 and 739.

- The amount ofnet bio-oil in the injections reported in the GHG Statement and the Isometric dMRV platform were slightly different than those obtained by reviewing the raw data including the weigh scale tickets for deliveries to Vaulted Deep of total injectate.
    - For example, the bio-oil mass for removal 732 was reported as 26.162 tonnes, while calculations based on the weigh scale tickets show that it is 26.190 tonnes.
    - These minor discrepancies are primarily traced to rounding errors in calculations and during migration of data among data platforms.
- Review of the transportation distance from AECN to El Dorado indicated a value of 3261 km vs a reported value of 3267 km – both from Google Maps using the same criteria.

Calculations of total net removals were completed based on use of the raw data values for both removal 732 and 739 and the revised distance. Based on this analysis, the total potential difference in net CO2 removal is 0.21%, equivalent to approximately 498 kg (0.498 tonnes) out of the total of 236.23 tonnes of net removals claimed in the GHG statement. **This not a material difference and the value of 236.23 tonnes is verified**.

5.2. VERIFICATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard and because of an outstanding non-conformity (see 5.4):

☐ ***Meets the requirements of the Isometric Standard V1.2.0, Biomass Geological Storage Protocol V1.0.1, Biomass Feedstock Accounting Module V1.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.0.2 for Verification***

☐ ***Meets the requirements with <u>minor revisions</u> of the Isometric Standard V1.2.0, Biomass Geological Storage Protocol V1.0.1, Biomass Feedstock Accounting Module V1.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.0.2 for Verification***

☒ ***Does Not Meet the requirements of the Isometric Standard V1.2.0, Biomass Geological Storage Protocol V1.0.1, Biomass Feedstock Accounting Module V1.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.0.2 for Verification***

5.3. VERIFICATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Biomass or Bio-oil Storage in Salt Caverns Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No change since validation
2.5.1	*Boundaries*	N	Acceptable – No change since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality - Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality - Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation
2.5.5	*Default Emission Factors, Proxies, and Models*	N	Acceptable – No change since validation. See findings (5.4) and recommendations (5.5) for forward action requests and recommended updates
2.5.6	*Common Calculation Factors*	N	Acceptable – No change since validation
2.5.7	*Uncertainty*	N	Acceptable – No change since validation
2.5.8	*Durability and Monitoring*	N	Acceptable – No change since validation
2.5.9	*Risk of Reversal*	N	Acceptable – No change since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	N	Acceptable – No change since validation
2.5.11	*Life Cycle Assessment (LCA) Policies*	N	Acceptable – No change since validation. LCA calculations updated for current reporting period.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	N	Acceptable – No change since validation
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Not Acceptable – Maintenance of well operation permits meets requirements and ongoing permit reporting requirements completed. However, the chloride concentration of injectate was exceeded for certain injections, which is not in compliance with the permit.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since validation
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since validation
3.0	*Counterfactual Definition*	N	Acceptable – No change since validation
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since validation
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since validation
Biomass or Bio-oil Storage in Salt Caverns Module			
1.0, 2.2	*Salt Cavern Characterization – Cavern Dimensions*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Integrity*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Porosity & permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Subsurface Features*	N	Acceptable – No change since validation
1.0	*Salt Cavern Characterization – Temperature, pH, conductivity/Cl- conc.*	Y (Required Monitoring Updated)	Not Acceptable – Brine characteristics measured prior to emplacement in accordance with the Class V Permit. Brine periodically tested and monitored as required and evidence updated. Brine concentrations (chloride) for multiple batches did not be the permit Cl level requirement for injections.
1.0	*Salt Cavern Characterization – Brine TOC*	Y (Required Monitoring Updated)	Acceptable – Brine TOC measured prior to emplacement, as required by Class V permit. Confirmed via updated TOC data.
1.0	*Salt Cavern Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – No change since validation
1.0, 3.1.3	*Salt Cavern Characterization – Surface Subsistence*	N	Acceptable – No change since validation
1.0	*Salt Cavern Characterization – Displaced Brine Injection location*	N	Acceptable – No change since validation
1.0, 2.3	*Emplacement Infrastructure*	N	Acceptable – No change since validation
1.0	*Emplacement Monitoring*	Y (Required Monitoring Updated)	Acceptable – monitoring completed and updated as required by Class V well permit
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
2.1	*Permit for Operation*	N	Acceptable – No change since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
3.1.1	*Injection Pressure*	Y (Required Monitoring Updated)	Acceptable – Pressure is continuously monitored and reported to KDHE monthly. Updates provided as required.
3.1.1	*Volume monitoring*	Y (Required Monitoring Updated)	Acceptable – Monitoring of injected material measured daily and reported monthly, confirmed by sonar quarterly. Updates provided as required.
3.1.1	*Injectate monitoring*	Y (Required Monitoring Updated)	Not Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required. However, monitoring for Cl indicates levels exceed permit levels for several injectate batches
3.1.1	*Sampling Frequency*	N.	Acceptable – No change since validation. Three samples collected for every injection batch. All sampling data provided and updated as required
3.1.1	*Safety Monitoring*	N	Acceptable – No change since validation
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since validation
3.1.3	*Well Pressure Monitoring*	Y (Required Monitoring Updated)	Acceptable – Well pressure measured continuously with safety interlock. Daily pressure reported to KDHE monthly. Updated monitoring data provided as required.
3.1.3	*Sonar Survey*	N	Acceptable – No change since validation
3.1.3	*Fill Volume*	Y (Required Monitoring Updated)	Acceptable – Cavern volume tracked based upon volume of emplaced material. Estimate volume reported monthly, verified by sonar quarterly. Updated monitoring data provided as required.
3.1.3	*Return Brine Monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation. New gas monitoring system currently being installed.
3.3	*Reversal Risk*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. VERIFICATION FINDINGS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed, except for one non-conformity, which remains and cannot be resolved at this time. This non-conformity relates to the chloride (salt) concentration of the injectate for several batches. For Charm Batch Numbers 200006 (rmv_1J9SN7T5R1S0D4D6), 200008 (rmv_1J9SF796D1S00FB4), 200011 (rmv_1J9XM7NHH1S0P8QJ), and 200013 (rmv_1J9XMAJFX1S0APY0) Charm did not meet the current Vaulted permit Chloride limit for injectate of 190,000 mg/l Chloride as specified in Section II.C-Table 3 of Vaulted's Class V permit. ***Therefore, although the values of injected carbon were verified, the project was not in compliance during the reporting period and is therefore not eligible for crediting in accordance with the applicable Isometric standards and protocols***.

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Forward Action requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

ID No	Type	Finding / Issue	Source	Protocol Reference	Action
16	Non-conformity	Chloride concentration for Charm Batch Numbers 200006 (rmv_1J9SN7T5R1S0D4D6), 200008 (rmv_1J9SF796D1S00FB4), 200011 (rmv_1J9XM7NHH1S0P8QJ), and 200013 (rmv_1J9XMAJFX1S0APY0) exceed the current Vaulted permit Chloride limit for injectate of 190,000 mg/l Chloride as specified in Section II.C-Table 3 of Vaulted's Class V permit		3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)	Incomplete. Evidence not available. Charm has provided a document explaining issues with chloride concentration limits due to solubility in non-aqueous bio-oil fraction to Vaulted for discussion with KDHC and requesting a permit modification.
1	Forward Action Request	Charm injection site (Vaulted Deep operated) is installing new gas monitoring equipment to better identify gas production at the wellhead / brine holding tanks. Updated information should be provided	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	Provide specifications, design, installation information, calibrations, and operational data for the new gas monitoring equipment when it becomes available and for relevant reporting periods
7	Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing.	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1		Please provide AECN testing report. Note, not applicable for this reporting period, but should be updated when possible
12	Forward Action Request	Review and update the KS grid electricity EF annually based on changes to grid mix and any eGrid updates			Please review and update as needed going forward
9	Forward Action Request	Biomass feedstock sourcing was confirmed at validation. Because of its critical nature for compliance, we feel it should be confirmed for each verification reporting period - demonstrating no change in feedstock sources since validation			Please provide confirmation of biomass feedstock sourcing and that it is not different from validated sources (mill wastes) for verifications in the future

6	Forward Action Request	Any activities associated with potential rail car cleanout and waste / residual bio-oil disposal in the future should be accounted for in LCA / /net removal calculations			For Charm-leased rail cars, monitor future operations to ensure any potential rail car cleanouts and waste disposals are identified if they occur and are accounted for in LCA
2	Recommendation	Transportation emissions utilize "the average "deadhead" or unloaded journey distance for a vehicle of this type according to USDA analysis". This is sufficient for accounting for return trips, but it may be preferable or more appropriate to use return trip distance if the trip route is known (i.e. a full return trip if applicable).		Bio-oil Geologic Storage 7.4.3 (Eq7)	Consider requesting return trip information from biomass delivery and bio-oil delivery transportation services and account for return trip accurately if appropriate

6. REVISION HISTORY

Original date of issue: November 8, 2024

Version	Date Issued	Noted Changes
Draft Version (v1.0)	Nov 8, 2024;	NA
Draft Version (v1.1)	Nov 15, 2024	Minor edits and comments from internal QA
Final Version (v1.2)	January 21, 2025	Minor edits and comments from Isometric and Charm. Closure of audit with remaining non-conformity.

References

[1] Isometric , *Isometric Standard, Version 1.4.0, Edition 12 Aug 2024.*
https://science.isometric.com/standard

[2] Isometric , *Biomass Geological Storage, Version 1.1, Edition 13 Sep 2024.*
https://science.isometric.com/protocol/biomass-geological-storage

[3] Isometric , *Biomass Feedstock Accounting, Version 1.2, Edition 21 May 2024.*
https://science.isometric.com/module/biomass-feedstock-accounting

[4] Isometric , *Biomass or Bio-oil Storage in Salt Caverns, Version 1.1, Edition 13 Sep 2024.*
https://registry.isometric.com/module/salt-cavern-storage

See Appendix 1 and Table 3 for list of specific files reviewed during the VVP.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

File Type	File Name
PDD	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf
GHG Statement	[External] May GHG Statement Report v1.1.docx
GHG Statement	Charm May 24 GHG Notes from Charm.pdf
GHG Statement Data from Isometric Platform	gss_1JAB46M2Y1S0H2KG.xlsx
Source Data	
Density and TAN Analytical Results	Charm MRV\200014 Density and TAN.pdf
Density and TAN Analytical Results	Charm MRV\200005 + 200006 + 200007 Density and TAN.pdf
Density and TAN Analytical Results	Charm MRV\200008 Density and TAN.pdf
Density and TAN Analytical Results	Charm MRV\200009 + 200010 + 200011 Density and TAN.pdf
Density and TAN Analytical Results	Charm MRV\200012 Density and TAN.pdf
Density and TAN Analytical Results	Charm MRV\200013 Density and TAN.pdf
Vaulted Salt (brine) delivery ticket	Vaulted (Salt) 05302024.png
Fuel purchases	Security Oil 05072024.pdf
Fuel purchases	Security Oil 05132024.pdf
Fuel purchases	Security Oil 05222024.pdf
Fuel purchases	Security Oil 05262024.pdf
Fuel purchases	Security Oil 05312024.pdf
Fuel purchases	Security Oil tLqV8w4.pdf
LCS purchases	Univar 05142024.PDF
LCS purchases	Univar 05222024.PDF
LCS purchases	Univar 05302024.PDF
Vaulted emplacement data logs	5_31 Emplacement 2.pdf
Vaulted emplacement data logs	5_10 Emplacement 1.pdf
Vaulted emplacement data logs	5_10 Emplacement 2.pdf
Vaulted emplacement data logs	5_11 Emplacement 1.pdf
Vaulted emplacement data logs	5_11 Emplacement 2.pdf
Vaulted emplacement data logs	5_16 Emplacement 1.pdf
Vaulted emplacement data logs	5_16 Emplacement 2.pdf
Vaulted emplacement data logs	5_16 Emplacement 3.pdf
Vaulted emplacement data logs	5_21 Emplacement 1.pdf
Vaulted emplacement data logs	5_21 Emplacement 2.pdf
Vaulted emplacement data logs	5_23 Emplacement 1.pdf
Vaulted emplacement data logs	5_23 Emplacement 2.pdf
Vaulted emplacement data logs	5_24 Emplacement 1.pdf
Vaulted emplacement data logs	5_25 Emplacement 1.pdf
Vaulted emplacement data logs	5_25 Emplacement 2.pdf
Vaulted emplacement data logs	5_28 Emplacement 1.pdf
Vaulted emplacement data logs	5_28 Emplacement 2.pdf
Vaulted emplacement data logs	5_30 Emplacement 1.pdf
Vaulted emplacement data logs	5_30 Emplacement 2.pdf
Vaulted emplacement data logs	5_31 Emplacement 1.pdf

Isometric dMRV Platform Source files (to include all weigh tickets/BOL, C content analyses, energy usage, transportation distances, embodied emissions factors, etc.)	ggs_1JAB40DHS1S09RW6-sources\src_1J09YMYVT1S0MYNE_5146_19a62478-6089-4fd1-9551-d5b0e273fe65.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YMZ8Z1S08KH3_5145_868e56f8-6294-4c69-9e44-15a1af848192.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YMZ9Y1S0FX8D_5148_8657192b-c2d7-4b63-83ad-992404eb9190.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YMZ981S0W8CR_5142_bfe71fcc-7ccf-412d-aecc-1b2a9431c5b0.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YMZA41S03J42_5141_a79c2d8b-c750-4856-be94-75afeb817cdc.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YMZAK1S0Q6ZQ_5149_da0e1f02-0edc-4987-8ac8-b01c8f5f4821.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YMZXZ1S0AVVC_5155_4ed52cdc-48c9-4757-91be-f43a4a8085c3.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YN0AX1S05TEB_5158_937cc492-ef23-43d4-a735-aa28d10242e2.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN0B01S0J5JP_5161_7e3dd154-4d6b-475f-986e-85b53f43fab0.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN0DP1S0SFA0_5160_1f32ad3d-b5cc-4b1d-860e-f97c2b80e86d.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN0ES1S0D45N_5157_4d902ee3-fec9-4ef5-8bf7-8795b75e1f5a.pdf ggs_1JAB40DHS1S09RW6-sources\src_1J09YN0S01S00S1A_5163_e8627383-4087-4cd8-bfe6-19e54ac87b3e.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN1ZR1S082RM_5168_2b450e91-c23e-4582-8731-71c8e469a3a9.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN2BT1S031BK_5169_948915e6-ed65-4315-915b-2c06dff796f8.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN2DE1S0PP78_5167_3e7d7358-eba5-45a9-af5d-942eff91943c.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN3SX1S0HMT7_5172_9f4a09b1-fcaf-45d3-9925-34d0d178b700.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN4ZG1S059NW_5174_d6bf1641-081f-41f5-96d2-6efca22c58a1.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN05Z1S0YGQ1_5154_cbf66a6d-e722-46fb-8a7e-31db5159eb67.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN5DA1S0CKD6_5175_05662faa-573a-4a2b-9905-ec2c7a303aa6.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN5ND1S0088V_5177_f16ed753-e8f9-49c4-a60e-3c2d3dd6f2fa.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN5V11S0KX4G_5178_ab60df12-f00d-47ed-88de-e3a3ad0c6a57.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN7ME1S09TAE_5185_36a3ab0a-a741-4a63-9445-95ace6117826.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN8B81S04RXD_5191_0c372a84-316f-4503-90b3-049ada1fce98.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN27V1S0FCFY_5165_f796a35d-761e-43a6-a503-4e43cf78667b.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN52X1S0RYHH_5173_fc48ab56-7f74-4728-be5f-3172f40278a5.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN60D1S07J05_5179_dbb6c258-6574-4341-a436-3991e83555b2.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN65E1S0V6VT_5183_3f464c46-3682-41a1-9019-0d6ed09a6c98.png ggs_1JAB40DHS1S09RW6-sources\src_1J09YN84B1S0H41R_5190_163275a4-c397-4dca-beb6-7df4a1795941.png

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APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Audit Evidence	Protocol Reference	Required Action	Conclusion / Resolution
1	Forward Action Request	Charm injection site (Vaulted Deep operated) is installing new gas monitoring equipment to better identify gas production at the wellhead / brine holding tanks. Updated information should be provided	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	3.1.3, 3.2 (Bio mass or Bio-oil Storage in Salt Caverns Module)	Provide specifications, design, installation information, calibrations, and operational data for the new gas monitoring equipment when it becomes available and for relevant reporting periods	Maintained as FAR. According to Charm, Vaulted has not yet installed this equipment. The requested information will be provided once installation is complete
2	Recommendation	Transportation emissions utilize "the average “deadhead” or unloaded journey distance for a vehicle of this type according to USDA analysis". This is sufficient for accounting for return trips, but it may be preferable or more appropriate to use return trip distance if the trip route is known (i.e. a full return trip if applicable).		Bio-oil Geologic Storage 7.4.3 (Eq7)	Consider requesting return trip information from biomass delivery and bio-oil delivery transportation services and account for return trip accurately if appropriate	Complete. Noted by Charm: “There is some uncertainty here as all of Charm's transport is handled by third-party vendors. We will assess where there is reasonable certainty to use actual routing in the future and do so on a case-by-case basis.”
3	Clarification	Salt and LCS addition records (raw data from operator logs) for lots 896 and 822 not available. Spreadsheets provide totals only.			Please provide or identify location of operator logs of salt and LCS additions for Lot 896 (removal 739) and Lot 822 (removal 732)	Complete. Records provided confirm Salt and LCS addition values.
4	Clarification (immaterial nonconformity)	Apparent rounding errors in transfer of raw data to Charm Ledger result in difficulty in tracing raw data to final removals. For example, for Removal 732, the bio-oil mass injected is documented as 26190 vs reported as 26120 kg			Provide snapshot of Ledger data showing the input injectate weigh scale values, salt mass, LCS mass and total bio-oil mass as compared to the raw data values for Removal 732 and 739	Complete. Discussion and demonstration of rounding examples demonstrates potential for small, non-material discrepancies.

5	Clarification	If 100% of Lot 820 and 821 are used to form Lot 822, which is the injection lot amount, There is a significant difference between the sum of Lot 820+821 = 31270kg vs Lot 822 documented weight (26120 kg). Is this the normal level of loss / residual from transfer? (Note that this does not matter for total removals values, since they are based on injection masses, but checking for completeness / understanding)			Please clarify	Complete. Explanation provided by Charm. Values are not losses, but total weight includes additions to bio-oil (salt and LCS), with some very small potential losses represented.
6	Clarification	Regarding the potential for residuals in transport vehicles above - where do those residuals end up (if they exist)? Are they drained and disposed somewhere? eventually pumped out and injected?			Please clarify	Converted to forward action request for Charm-leased rail cars to monitor any cleaning and waste disposal required and accounting in LCA for future verifications.
7	Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing. Has this been completed yet?	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1		Please provide AECN testing report. Note, likely not applicable for this reporting period, but should be updated when possible	Testing not completed yet. Remains as FAR
8	Clarification	Current AECN pyrolyzer stack emissions model only addresses CH4 and N2O emissions. There are likely CO2 emissions also - are these netted out as a result of the use of biomass on a 1:1 basis?	src_1J09YMZA41S03J42_5141_a79c2d8b-c750-4856-be94-75afeb817cdc.pdf		Please clarify	Complete. Confirmed in compliance with 7.4.3.5 of Isometric Bio-Oil Geologic Storage Protocol.

9	Clarification / Forward Action Request	Biomass feedstock sourcing was confirmed at validation. Because of its critical nature for compliance, we feel it should be confirmed for each verification reporting period - demonstrating no change in feedstock sources since validation			Please provide confirmation of biomass feedstock sourcing and that it is not different from validated sources (mill wastes)	Included as FAR for future verifications
10	Clarification	Isometric platform states reporting period of April 30 - May 30 2024 vs GHG statement stating May 1 - May 31, 2024	[External] May GHG Statement Report v1.1.docx gss_1JAB46M2Y1S 0H2KG.xlsx		Please clarify and update GHG Statement (or Isometric platform) as needed	Complete. Reporting period is Ma1-May 31 2024
11	Clarification	There is no documentation of any reporting / accounting of any releases or spills of bio-oil throughout the reporting period. We assume that no report = no spills. Note that this would only impact emissions associated with containment and cleanup/disposal, not total gross CDR	Bio-oil Geologic Storage 7.4.1.3		Please confirm.	Complete. No spills or release confirmed by Charm.
12	Forward Action Request	Review and update the KS grid electricity EF annually based on changes to grid mix and any eGrid updates			Please review and update as needed going forward	Maintain as FAR – confirmed by Charm for review at next eGrid revision.
13	Nonconformity	Please provide updated tracking log of all Vaulted permit and reporting compliance logs that may be required to be updated to confirm ongoing permit compliance. This includes the quarterly cavern fill and emplacement reports. Please provide copy of submitted report as evidence (Should be April 2024 report for the reporting period of interest	Bio-oil Geologic Storage 3.1		Please provide	Complete. All logs provided and compliance demonstrated

14	Nonconformity	For well monitoring, please provide summary of well cavern pressure and T monitoring (continuous / daily required for P and 6 mos for T), Wellhead gas monitoring (monthly) and	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)		Please provide	Complete. All logs provided and compliance demonstrated
15	Nonconformity	Where are conductivity tests of injectate documented? Vaulted injectate sheet indicates Charm to provide.	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)		Please provide	Complete. Documents provided by Charm and confirmed compliant.
16	Nonconformity	Chloride concentration for Charm Batch Numbers 200006 (rmv_1J9SN7T5R1S0D4D6), 200008 (rmv_1J9SF796D1S00FB4), 200011 (rmv_1J9XM7NHH1S0P8QJ), and 200013 (rmv_1J9XMAJFX1S0APY0) exceed the current Vaulted permit Chloride limit for injectate of 190,000 mg/l Chloride as specified in Section II.C-Table 3 of Vaulted's Class V permit	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)		Please provide evidence that required permit conditions were met	Incomplete. Evidence not available. Charm has provided a document explaining issues with chloride concentration limits due to solubility in non-aqueous bio-oil fraction to Vaulted for discussion with KDHC and requesting a permit modification.

APPENDIX 3: ISOMETRIC STANDARD & PROTOCOL CHECKLISTS

APPENDIX 4: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	Criteria Met?	Evidence / Notes *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Lily Schacht
Carbon Removal Verification Engineer, 350Solutions

EDUCATION:
MS, Environmental Engineering, University of Wisconsin – Madison, 2019
BS, Chemical, Energy, and Environmental Engineering, Washington University in St. Louis, 2017

EXPERIENCE SUMMARY:
Lily Schacht is an Environmental and Chemical Engineer with experience in process engineering, environmental chemistry, analytical methods, and life cycle analysis (LCA). At 350Solutions, Lily works on verifying carbon dioxide removal (CDR) technologies, with a focus on mineralization-based pathways, including enhanced weathering, direct air capture, and ocean alkalinity enhancement. Previously, Lily led agronomic research at an enhanced weathering CDR supplier where she organized field trials across multiple states to quantify carbon removal and crop yield changes after rock application. Before that, Lily developed rapid prototyping instrumentation to optimize a biomineralization process in concrete production and aided in scaling up the process to pilot-scale. Lily also built environmental impact models for process variable sensitivity analysis of demo-scale manufacturing processes. These models were used to drive the direction of research and development to minimize product life cycle impacts. Throughout her career, Lily has developed analytical chemistry methodologies for both liquid- and solid-phase analyses.

RESEARCH AND PROFESSIONAL EXPERIENCE:
March 2024 – Present: Carbon Removal Verification Engineer, 350Solutions
Verify CDR technologies on behalf of registries and the XPRIZE Carbon Removal challenge. Specializes in mineralization pathways, including mineralization kinetics, measurement methods, and open-system modeling.

Nov 2022 – Feb 2024: Researcher, Lithos Carbon
Quantified carbon removal rates and agronomic impacts of enhanced weathering on cropland across six US states; Evaluated chemical analysis methods for precision relative to cost.

Nov 2023 – Jan 2024: Independent Consultant, Keel Labs
Built an environmental impact model to evaluate potential material and process changes; Recommended areas for reducing material usage up to 80%

Dec 2020 – Oct 2022: Research Scientist II
Guided experimentation and data analysis throughout all R&D teams to inform techno-economic analysis (TEA) and LCA; built and led the carbonate biomineralization prototyping workstream for rapid iteration; developed real-time measurement techniques for critical process parameters in solid state

SELECTED PUBLICATIONS & PRESENTATIONS:

- **Schacht, L.**, Baum, M., Liu, H., & Yap, M. (2023) Scaling Enhanced Rock Weathering: Agronomic Impacts at Field-Scale [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO.
- **Schacht, L.** and Ginder-Vogel, M. Arsenite Depletion by Manganese Oxides: A Case Study on the Limitations of Observed First Order Rate Constants. *Soil Syst.* 2018, 2(3), 39. https://doi.org/10.3390/soilsystems2030039

350Solutions, Inc. Corporate Experience

350Solutions serves as an independent expert in cleantech, low carbon, and environmental technologies. We provide an unbiased assessment of innovative technologies. 350Solutions is accredited through ANAB under ISO 17020 as an independent inspection body to provide independent technology evaluation services using the ISO 14034 ETV process. 350Solutions is an approver verifier for both the Isometric and Puro. Eart carbon removal registries. In addition, 350Solutions staff include a Certified Measurement and Verification Professional (CMVP for IPMVP) and a North Carolina Registered Professional Engineer (P.E.).